



Q1: (Please write your answers in a table only; otherwise, your answers will be neglected) (22 marks)

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	B	B	A	C	A	A	B	A
Question	11	12	13	14	15					
Answer	B	A	A	B	C					

+21

1. We cannot send the digital signal directly to the channel when the channel is:

- A. bandpass
- B. Bypass
- C. baseband
- D. Low pass

2. The transmission impairment refers to a signal with high energy in a very short time is

- A. Thermal Noise
- B. Induced Noise
- C. Cross-talk
- D. Impulse Noise

3. A digital signal is a composite analog signal with

- A. finite bandwidth
- B. Infinite bandwidth
- C. zero bandwidth
- D. None of the above

4. In data communications, the non-periodic signals

- A. Sine wave
- B. Digital Signals
- C. Analog Signals with T period
- D. None of the above

5. A communication pathway that transfers data from one point to another is called

- A. Link
- B. Node
- C. Medium
- D. Topology



6. In a bus topology, the device linking all the nodes in a network is

- A. Router
- B. DropLine
- C. Backbone
- D. hub

7. Which topology covers security, robust, and eliminates the traffic factor?

- A. Mesh
- B. Ring
- C. Star
- D. Bus

8. The bus, ring, and star topologies are mostly used in the

- A. LAN
- B. MAN
- C. WAN
- D. Internetwork

9. The repeater is used in

- A. Mesh Topology
- B. Ring Topology
- C. Bus Topology
- D. Star Topology

10. In a human voice:

- A. The number of frequencies is infinite, but the range is limited
- B. The number of frequencies is finite, but the range is limited
- C. The number of frequencies is infinite, but the range is unlimited

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11. An amplifier has an output of 20 w. What is its output in dBw:
A. 13 dBmw B. 13 dBw C. 43 dBw D. 43 dBmw
12. The ideal channel has the value of SNR equal to:
A. ∞ B. 0 C. 1 D. $\frac{1}{2}$
13. The domain of duty of the top 3 layers in TCP/IP is:
A. internet B. link C. message
14. Connecting your cellular phone to the router at home makes a network, the phone is:
A. hub B. host C. switch

15. Bit length is measured by:

A. bits

B. bits/s

C. meters

$$dB = 10 \log_{10} 20$$

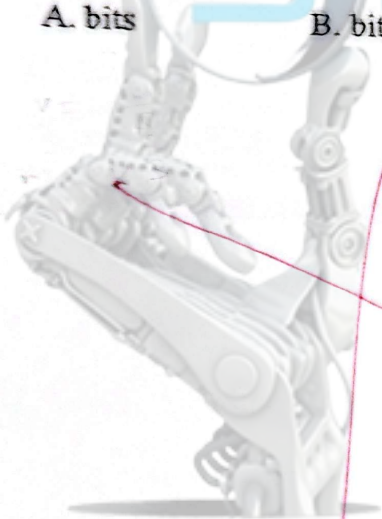
$$dB = 10 \log_{10} 20 = 13$$

$$\frac{1}{0} = \infty$$

M E C H A T R O N I C S

S P E C T R U M

T E A M



Q2: Suppose that a digitized TV picture is to be transmitted from a source that uses a 480×500 matrix of picture elements (pixels), where each pixel can take on one of 32 intensity values (5 bits/pixel). Assume that 30 pictures are sent per second. Find the source rate R (bps). Assume that the TV picture is to be transmitted over a channel with 4.5-MHz bandwidth and a 35-dB signal-to-noise ratio. Determine the channel capacity (bps). (8 marks)

$$\text{bps} = (480 \times 500) \text{ pixels/s} \times 5 \text{ (bit/pixel)} \times 30 \text{ pictures/s}$$

$$\text{bps} = 480 \times 500 \times 5 \times 30 = \boxed{36 \text{ Mbps}}$$

$$36 \times 10^6 \text{ bps}$$

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

$$= 4.5 \text{ MHz} \times \log_2(1 + 3162)$$

$$\text{Capacity} = \boxed{52.32 \text{ Mbps}}$$

$$= 52321850 \text{ bps}$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR})$$

$$35 = 10 \log_{10}(\text{SNR})$$

$$3.5 = \log_{10}(\text{SNR})$$

$$\text{SNR} = (3162.3)$$